Software Test Plan (STP)

IEEE-style test plan with test cases and traceability

Project: Pharmacy Expiry & Re-order Dispatch Engine

Version: 1.0 | Status: Student Submission Draft | Date: 12 October 2026

Prepared for: Software Engineering Mini-Project Deliverables – Part 1

Revision History

Version	Date	Author	Change
1.0	12-Oct-2026	Student	Initial submission draft

1. Introduction

1.1 Purpose

This test plan defines the scope, strategy, environment, schedule, responsibilities, and test cases for the Pharmacy Expiry & Re-order Dispatch Engine.

1.2 Scope

Testing covers batch entry, inventory display, FEFO eligibility and ordering, reorder-threshold alerts, purchase-order creation, performance, and role-based access. Live supplier fulfillment and clinical/pharmacy decision-making are outside this test scope.

1.3 References and Definitions

References: project SRS v1.0 and Software Architecture & Design Specification v1.0. FEFO means First-Expiry-First-Out; RTM means Requirements Traceability Matrix.

2. Test Items

- Batch Management
- Inventory View
- FEFO Selector
- Reorder Monitor
- Purchase Order Service
- Authentication and Authorization
- Inventory search/list performance

3. Features to be Tested

Requirement ID	Feature	Test focus
FR-01	Record medicine batch	Valid batch accepted; invalid fields rejected.
FR-02	View inventory	Correct persisted quantities and expiry labels.
FR-03	FEFO list	Ascending expiry order; expired/zero-stock excluded.
FR-04	Reorder monitoring	Below-threshold stock flagged correctly.
FR-05	Purchase order	Unique order and required fields/status.
NFR-01	Performance	95% of inventory/FEFO requests within 2 seconds at 5,000 records.
NFR-02 / SEC-02	Authorization	Unauthenticated and unauthorized actions denied.

4. Features Not to be Tested

- External supplier systems and actual delivery/fulfilment.
- Third-party identity provider internals, if one is used.
- Production-grade pharmacy regulatory certification; this is a student prototype.
- Clinical appropriateness of dispensing decisions.

5. Test Approach / Strategy

Levels: unit tests for validation and FEFO ordering; integration tests for services and database persistence; system tests for end-to-end inventory-to-order flows; acceptance tests with representative pharmacy workflows.

Types: functional, boundary-value, negative, regression, performance, usability, and security/authorization testing. Test data shall be synthetic and must not contain real patient data.

Entry criteria: stable build available; schema and test data prepared; test environment configured. Exit criteria: all planned test cases executed; no open critical defects; all high-priority requirements pass or have documented disposition.

5.1 Security Validation

- Verify unauthorized users cannot create or edit batches or purchase orders.
- Verify role restrictions for threshold configuration and order dispatch.
- Test malformed IDs, dates, quantities, and threshold values.
- Inspect logs to ensure credentials/secrets are not recorded.
- Verify audit records contain actor, timestamp, action, and record ID without sensitive values.

6. Test Environment

Hardware: developer laptop or agreed lab workstation. Software: application build, supported Python/runtime and database versions as defined by implementation, test database populated with synthetic data. Tools: pytest or equivalent, API client/browser as applicable, and a timer or load-testing tool for performance checks.

7. Test Schedule

Activity	Planned window	Output
Test design	Day 1	Test cases and data
Environment/data setup	Day 1	Ready test environment
Execution and defect logging	Days 2–3	Execution results and defects
Regression and summary	Day 4	Final test summary

8. Test Deliverables

- Software Test Plan
- Test cases and synthetic test data
- Execution evidence/logs
- Defect records
- Requirements Traceability Matrix
- Test summary report

9. Roles and Responsibilities

Role	Responsibility
Student / Test Engineer	Design and execute tests; capture results and defects.
Developer	Review failures, fix defects, and support retesting.
Project evaluator / Product Owner	Review requirement coverage and accept the submission.

10. Risks and Mitigation

Risk	Mitigation
Test data does not cover expiry edge cases	Include expired, expiring-soon, zero-stock, and equal-expiry records.
Database/environment unavailable	Keep a reproducible setup and resettable synthetic fixtures.
Performance varies by machine	Record hardware/configuration and repeat the same workload.

Authorization gaps	Test both allowed and forbidden roles for each privileged action.

11. Assumptions and Dependencies

- Requirements and acceptance criteria in SRS v1.0 are the baseline.
- Synthetic test accounts and medicine batches are available.
- The test database can be reset between cases.
- Supplier dispatch is tested as a system-recorded action unless an integration exists.

12. Suspension and Resumption Criteria

Suspend if the application cannot start, the test database is unavailable, or more than 30% of planned cases are blocked. Resume after the blocking defect is fixed and the environment is stable.

13. Test Cases and Traceability

Test ID	Traceability	Scenario	Test steps / data	Expected result	Result
TC-01	FR-01	Record valid batch	Enter valid medicine ID, batch ID, future expiry, quantity 20.	Batch is saved and visible.	Pass/Fail
TC-02	FR-01	Reject invalid quantity/date	Submit quantity -1 and then an invalid expiry date.	Each invalid submission is rejected with a field-level error; no invalid record is saved.	Pass/Fail
TC-03	FR-02	Inventory matches stored data	Open inventory for a known batch with quantity 20.	Medicine ID, batch ID, quantity, and expiry status match database values.	Pass/Fail
TC-04	FR-03	FEFO ordering	Create two eligible batches with different expiry dates.	Earlier expiry batch appears before later expiry batch.	Pass/Fail
TC-05	FR-03	Exclude expired and zero stock	Create one expired batch, one zero-stock batch, and one valid positive-stock batch.	Only the valid positive-stock batch appears in the FEFO list.	Pass/Fail
TC-06	FR-04	Low-stock threshold boundary	Set threshold 10; test available stock 9, 10, and 11.	9 is flagged; 10 and 11 are not flagged when rule is strictly below threshold.	Pass/Fail
TC-07	FR-05	Create purchase order	Manager creates order for flagged medicine with valid supplier and positive quantity.	Unique order ID is created and initial status is recorded.	Pass/Fail
TC-08	FR-05 / SEC-02	Reject unauthorized order creation	Attempt purchase-order creation as unauthenticated user or clerk without permission.	Request is denied; no order is created; event is logged without secrets.	Pass/Fail

TC-09	NFR-01	Performance check	With 5,000 batch records, execute 100 inventory searches/FEFO list requests.	At least 95 requests complete within 2 seconds on documented test machine.	Pass/Fail
TC-10	NFR-02 / SEC-01	Input and log security	Submit malformed IDs and inspect application logs.	Invalid input is rejected; logs contain no credentials, tokens, or unnecessary sensitive data.	Pass/Fail

14. Test Metrics and Reporting

Report execution percentage, pass/fail percentage, defects by severity, defect age, and requirement coverage. A requirement is covered when at least one linked test has been executed.

15. Approvals

Role	Name	Signature / Date
Student / Test Engineer		
Developer / Reviewer		
Project Evaluator		